

FTGP Group 11: Sprint Report 1

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1 Team Members

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- Fuyu Cao (ld25515)
- Tianwei Yu (np25024)
- Detai Dong (mz25325)

2 Product Vision Summary

EcoPassEU is a privacy-preserving Digital Product Passport (DPP) DApp designed for sustainable consumer goods in Europe, with an initial focus on textiles and small electronics. By scanning a QR code or NFC tag, users access verified product provenance, material composition, and repair/recycling guidance.

The project utilises a hybrid architecture: sensitive documents are stored off-chain, whilst cryptographic hashes, timestamps, and ownership proofs are maintained on-chain. This approach guarantees traceability and regulatory compliance without compromising commercial confidentiality, delivering a practical prototype for SMEs, consumers, and regulators.

3 Product Backlog

The current product backlog consists of the following foundational tasks:

- Define MVP scope and core user stories.
- Design the hybrid system architecture (on-chain and off-chain components).
- Set up the GitHub repository, branching strategy, and the Jira board.
- Design the product passport data structure for textiles and electronics.
- Build the initial Solidity smart contract skeleton for record creation.
- Explore and test IPFS-based off-chain document storage.
- Create the React frontend structure and core pages.
- Prototype a QR-code-based passport lookup flow.
- Prepare project documentation, sprint reports, and presentation materials.
- Review GDPR compliance and data immutability constraints.

Table 1. Sprint 1 Task Allocation and Details

Task	Description	Owner(s)	Due Date
Confirm MVP scope & user stories	Clarify prototype inclusions/exclusions. Identify manufacturer, consumer, and regulator roles.	Luxiao Cao	17th Mar
Set up GitHub & Jira board	Create shared repository, define branching/issues, and generate Sprint 1 backlog on Jira.	Detai Dong	17th Mar
Design system architecture	Produce diagram showing blockchain, IPFS, frontend, and QR/NFC. Agree on on/off-chain data split.	Akshansh Rajora	18th Mar
Define product data model	Decide core fields (ID, materials, provenance, repair/recycling). Prepare sample test records.	Tianwei Yu	19th Mar
Set up smart contract environment	Initialise Solidity/Hardhat project. Create skeleton for batch registration and metadata hash storage.	Fuyu Cao	20th Mar
Test IPFS off-chain storage	Upload sample JSON to IPFS, retrieve CID/hash, and verify smart contract linkage.	Akshansh Rajora	21st Mar
Create frontend scaffold	Set up React structure. Build Home, Passport View, and Issuer Upload Form placeholder pages.	Tianwei Yu	21st Mar
Prototype QR lookup flow	Generate sample QR code linked to a mock passport page and verify readability upon scanning.	Luxiao Cao	22nd Mar
Prepare sprint documentation	Record planning outcomes, update the report draft, and organise notes for the weekly review.	Fuyu Cao, Detai Dong	23rd Mar

4 Sprint 1 Planning (16th – 23rd March)

Table 1 comprehensively outlines the tasks, detailed descriptions, owners, and due dates for Sprint 1.

5 GitHub Repository and Jira Project Board

All group members have successfully joined the collaborative platforms required to manage this project via Scrum:

- **GitHub Repository:** https://github.com/DeTaiDong/FTGP2425_Group1_11.git
- **Jira Project Board:** <https://laurencecao.atlassian.net/jira/software/projects/SCRUM/summary>